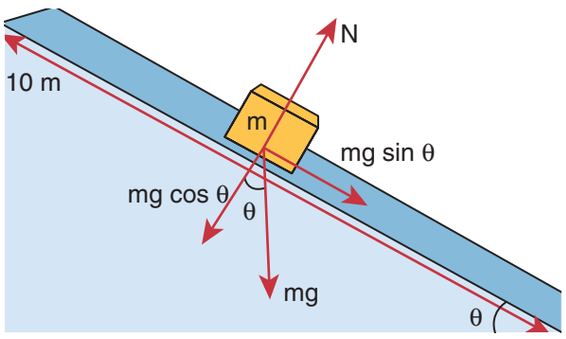
EXAMPLE 4.3

An object of mass m=1 kg is sliding from top to bottom in the frictionless inclined plane of inclination angle $\theta = 30°$ and the length of inclined plane is 10 m as shown in the figure. Calculate the work done by gravitational force and normal force on the object. Assume acceleration due to gravity, g = 10 m s^{-2}



Solution

We calculated in the previous chapter that the acceleration experienced by the object in the inclined plane as $g\sin\theta$.

According to Newton's second law, the force acting on the mass along the inclined plane $F = mg\sin\theta$. Note that this force is constant throughout the motion of the mass.

The work done by the parallel component of gravitational force $(mg\ \sin\theta)$ is given by

$$W = \vec{F}.d\vec{r} = Fdr\cos\phi$$

where ϕ is the angle between the force $(mg\ \sin\theta)$ and the direction of motion (dr). In this case, force $(mg\ \sin\theta)$ and the displacement $(d\vec{r})$ are in the same direction. Hence $\phi = 0$ and $\cos\phi = 1$

$$W = F\ dr = (mg\ \sin\theta)\ (dr)$$

(dr = length of the inclined place)

$$W = 1 \times 10 \times \sin(30°) \times 10 = 100 \times \frac{1}{2} = 50\ J$$

The component $mg\cos\theta$ and the normal force N are perpendicular to the direction of motion of the object, so they do not perform any work.

EXAMPLE 4.4

If an object of mass 2 kg is thrown up from the ground reaches a height of 5 m and falls back to the Earth (neglect the air resistance). Calculate

(a) The work done by gravity when the object reaches 5 m height
(b) The work done by gravity when the object comes back to Earth
(c) Total work done by gravity both in upward and downward motion and mention the physical significance of the result.

Solution

When the object goes up, the displacement points in the upward direction whereas the gravitational force acting on the object points in downward direction. Therefore, the angle between gravitational force and displacement of the object is 180°.

(a) The work done by gravitational force in the upward motion.

Given that $dr = 5m$ and $F = mg$

$$W_{up} = Fdr\ cos\theta = mgdr\ \cos 180°$$

$$W_{up} = 2 \times 10 \times 5 \times (-1) = -100 \text{ joule.}$$

$[\cos 180° = -1]$

UNIT-4(XI-Physics_Vol-1).indd 171 05-01-2022 22:19:33